

Developing FPGA Test Procedures for Memory Controller

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Introduction

- ICASL group produces integrated circuits (IC's) with the goal of withstanding high TID (total ionizing dose) and cryogenic temperatures for NASA's missions to Europa
- FPGA are configurable integrated circuits that we can use to mimic the behavior of the memory controller to validate testing procedures and performance

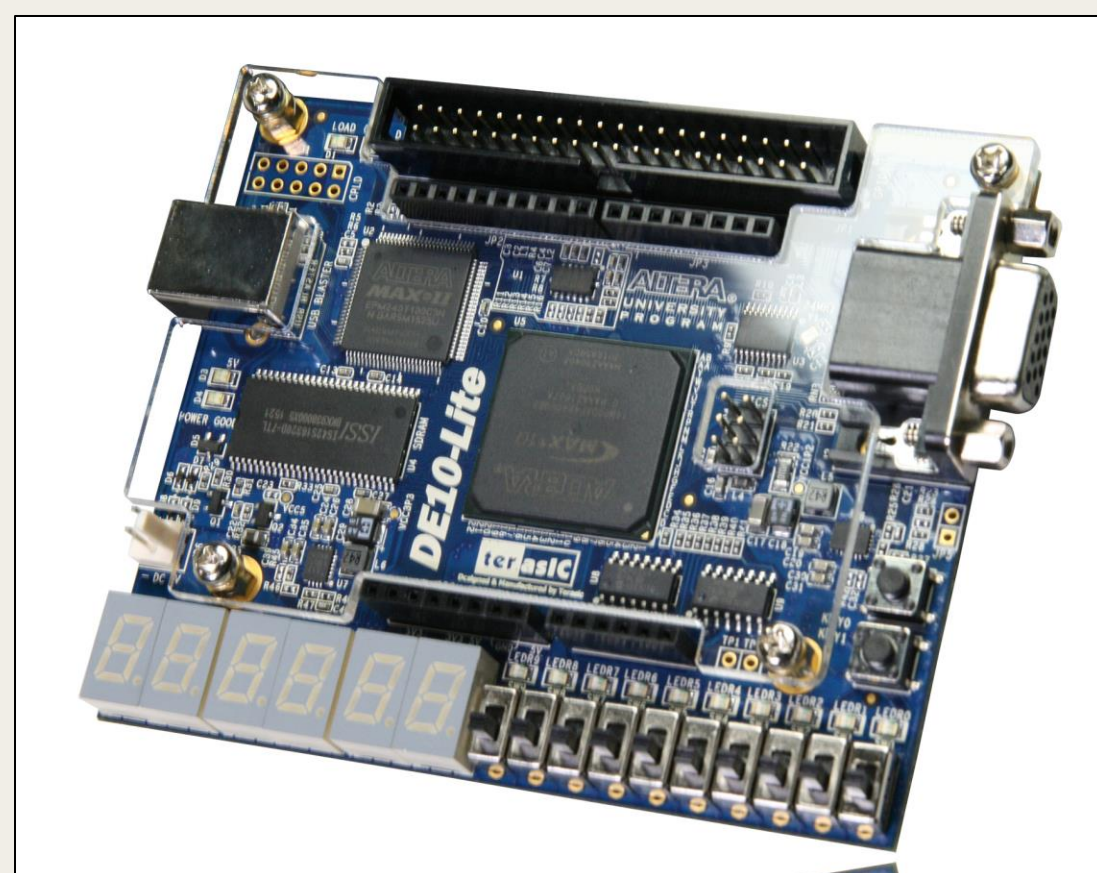


Figure 1. DE10-Lite Intel FPGA Development Board 64MB SDRAM [Image by Terasics Inc, DE10-Lite Board]



Figure 2. 64MB SDRAM [Image by TME Electronic Components, IS42S32200L-7TLISSI]

- Research Purpose: Field-Programmable Gate Array (FPGA) allows us to prototype an SDRAM controller and help with testing of in high radiation environments

Methodology

- Establishing communication between the CPU and SDRAM is essential and is done utilizing a proprietary memory controller from Jet Propulsion Laboratory
- The provided memory controller is pipelined and optimized to the IO count of ICASL's integrated circuit
- The memory controller is essential in managing data flow to enable read and write operations to memory
- Program FPGA through pin assignments to configure inputs as switches and display outputs by LEDs

Methodology Continued

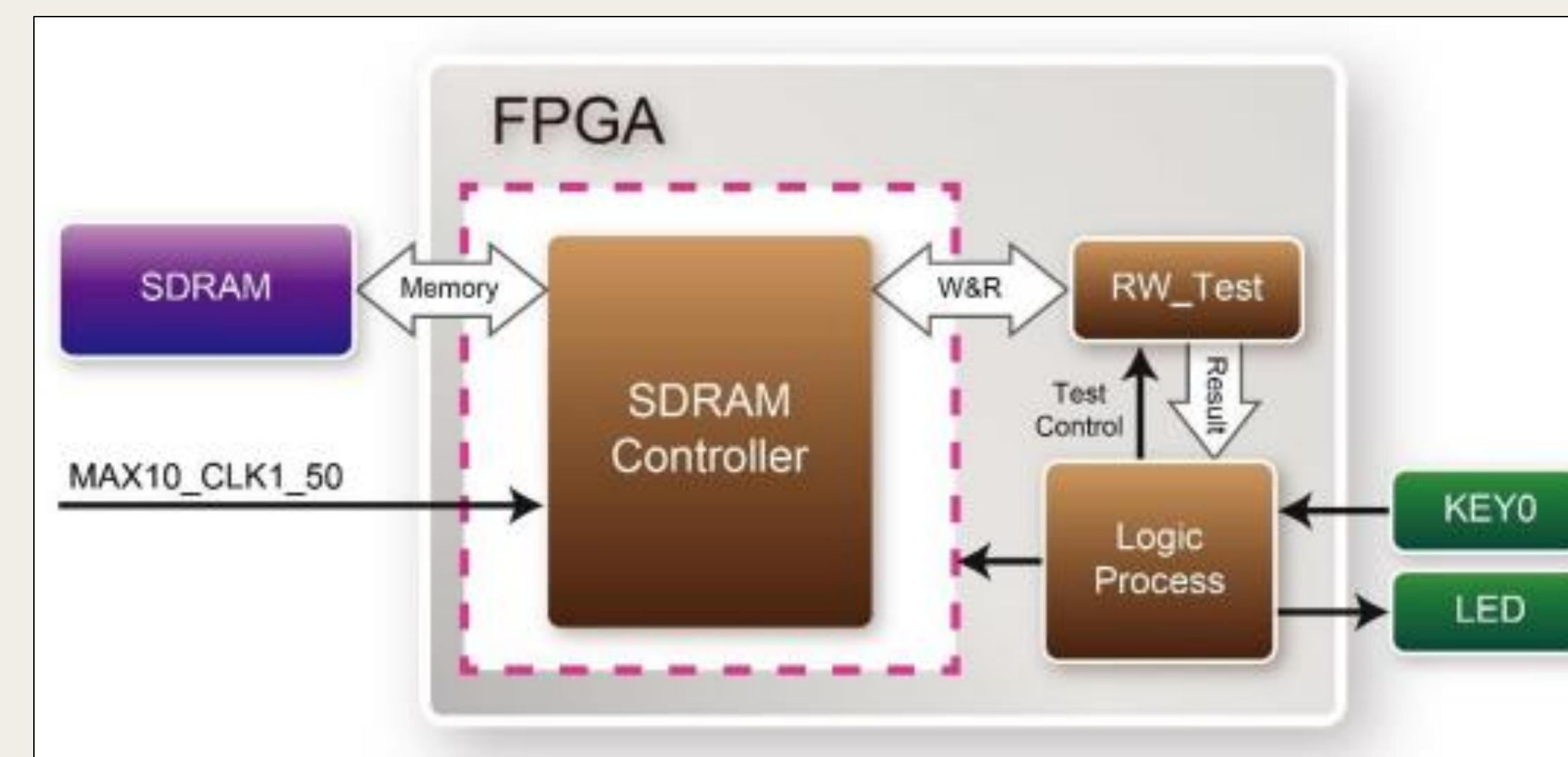


Figure 3. FPGA and Memory Controller Design [Image by Terasics Inc, DE10-lite User Manual]

- Initialized signal starts FPGA
- Grant output allows read and write commands
- Write input saves data to address
- Write strobe output validates data
- Read input reads data from address.
- Read strobe validates data
- Finished output declares completion of operations

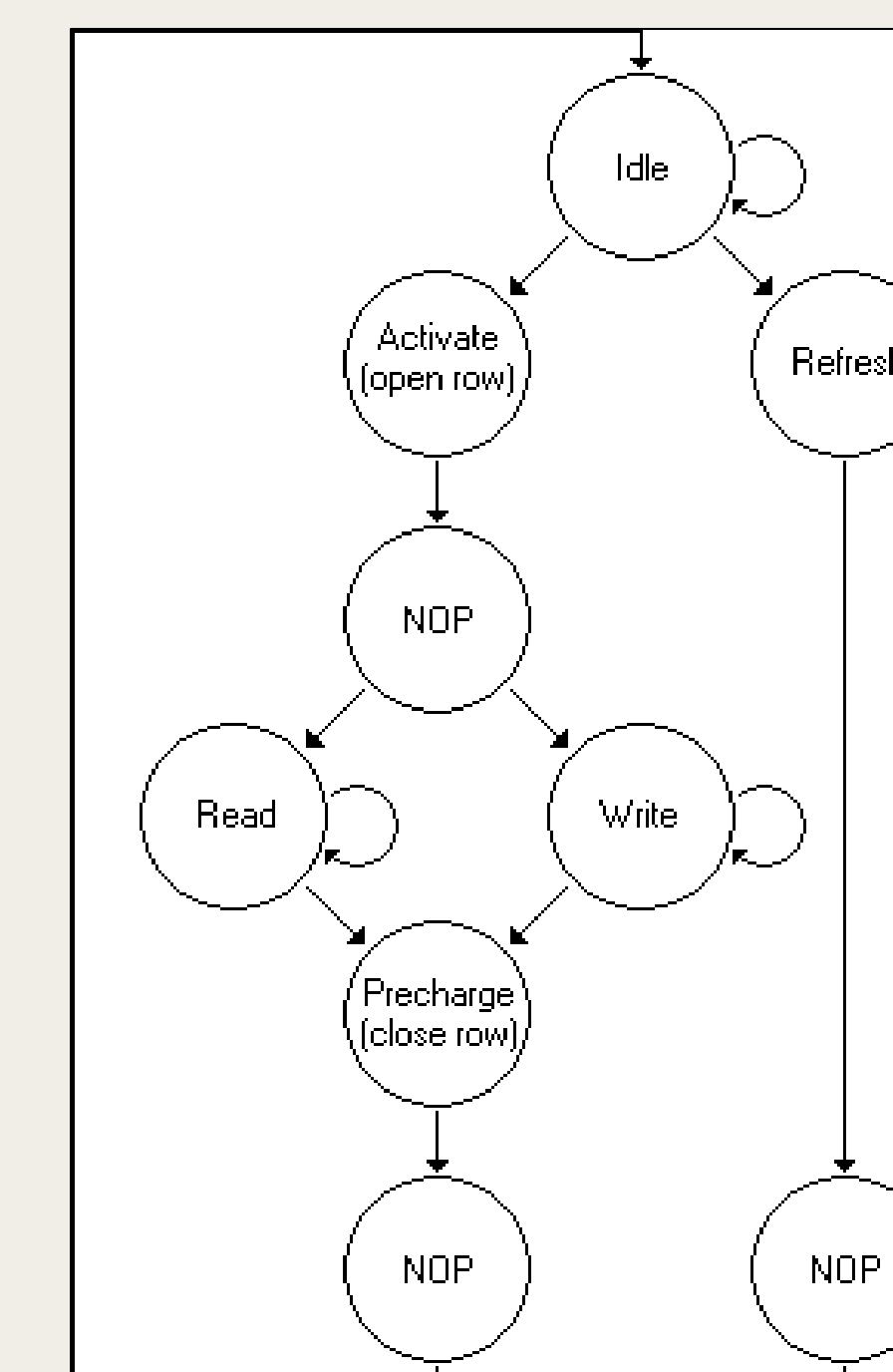


Figure 4. State Machine [Image by J P. Nicolle, fpga4fun, <https://www.fpga4fun.com/SDRAM2.html>]

Results

- Synthesize and simulation of RTL memory controller from Jet Propulsion Laboratory

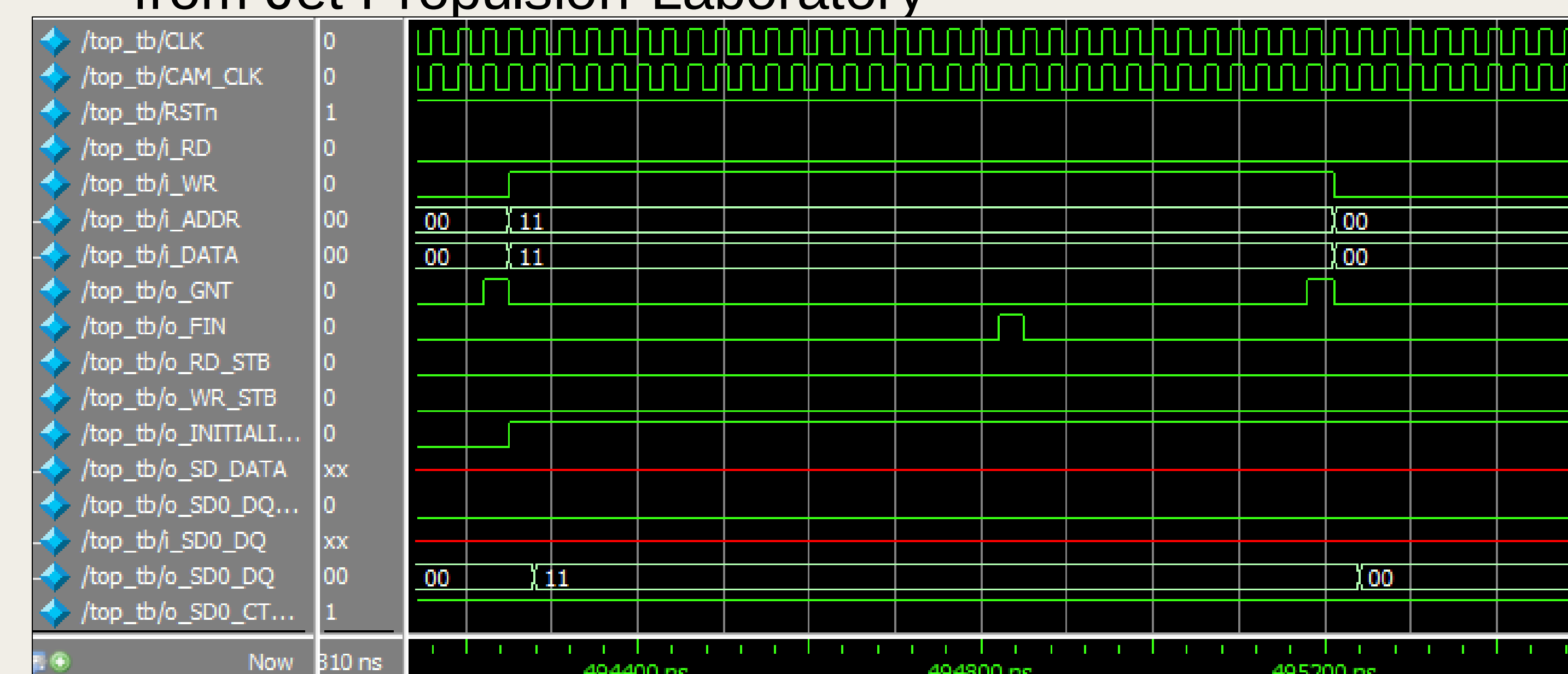


Figure 5. QuestaSim Testbench Simulation

Results Continued

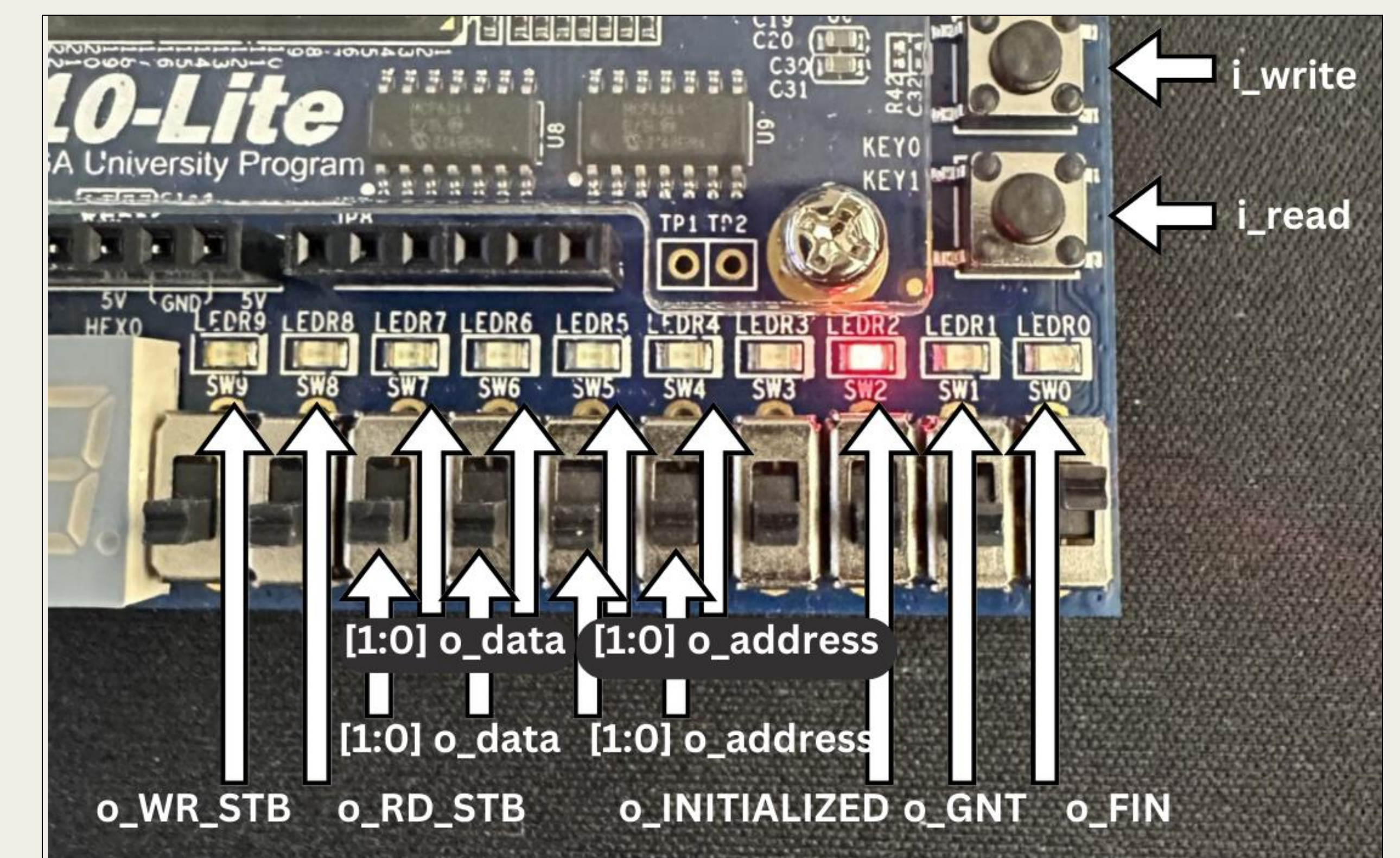


Figure 6. FPGA initialization

- Implementation of memory controller on FPGA
- Received initialization, grant (dim), and finish (dim) outputs
- Unable to obtain read and write strobe outputs to verify operations

Conclusion

- While FPGA initializes, it lacks proper reading and writing response
- Further analysis into memory controller and pin assignments will need to be completed to verify operations
- My project has led to the exploration of FPGA to validate integrated circuits design specific to ICASL's collaboration with NASA's mission to Europa.

References

1. Nicolle, J. P. (n.d.). SDRAM 2 - A simple controller. <https://www.fpga4fun.com/SDRAM2.html>
2. Terasics Inc. (2020, June 5). DE10-Lite User Manual. Hsinchu City, Taiwan.